Program: **Diploma in Electrical Engineering& Electrical Vehicle Technology**	
Course Code: **5411**	Course Title: **Electric Vehicle Technology & Design Considerations**
Semester**: 5**	Credits**: 4**
Course Category: **Program Core**	
Periods per week: 4(L:3 T:1 P:0)	**Periods per semester: 60**

Course Objectives:

- To understand general aspects of Electric and Hybrid Vehicles
- To understand the control system basics of EV's
- To understand basic design considerations of EV's
- To understand the basic concepts of braking systems and sensors

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic knowledge about storage batteries	2411	Introduction to Electric Vehicles &Elementary Theory	2
Basic Knowledge of batteries and charging systems	3411	Electric Vehicle Batteries & Charging Systems	3

Course outcomes:

On Completion of this course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Outline the Electric Vehicle and Hybrid Vehicle.	15	Understanding
CO2	Identify the various drive trains in EV & HEV	15	Understanding
CO3	Outline the basic design considerations of EV	15	Remembering
CO4	Outline the braking system for EV’s	13	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Outline the Electric Vehicle and Hybrid Vehicle.**		
M1.01	Explain the need for hybrid and electric vehicles	4	Understanding
M1.02	Explain the components and working principles of electric and hybrid vehicles.	4	Understanding
M1.03	Identify the various configurations of Electric Vehicles	3	Understanding
M1.04	Identify the types of hybrid Electric vehicle	4	Understanding
Contents: Need for hybrid and electric vehicles, Historical background. Social and environmental importance of hybrid and electric vehicles. Main components and working principles of electric and hybrid vehicles. Different configurations of Electric Vehicles (EV) – Pure Electric vehicle (PE)-Battery Electric vehicles (BEV)- Fuel Cell Electric Vehicles (FCEV) -Hybrid Electric vehicle (HEV) – Plug-in Hybrid Electric vehicle (PHEV) - Types of Electric vehicles based on with and Without differential. Types of HEV - series, parallel, series-parallel and Complex. Basic idea of EREV (Extended range Electric Vehicle)			
CO2	**Identify the various control systems & drive trains in EV & HEV**		
M2.01	Explain the control systems for EV’s	4	Understanding

M2.02	Explain the split power devices for HEV	3	Understanding
M2.03	Identify various drive trains in Electric Vehicles and Hybrid Electric Vehicles	4	Understanding
M2.04	Identify the braking systems in EV	4	Understanding
	Series Test I	1	

Contents:

Control systems for EV - Sizing the drive system - Sizing the propulsion motor, sizing the power electronics - Selecting the energy storage technology - Communications - supporting subsystems. Power split devices for Hybrid Vehicles.
Drive train in EV and HEV: Basic Architecture of Electric Drive Trains: General configuration of an electric vehicle, Alternatives Based on Drive train Configuration, Alternatives Based on Power Source Configuration, Single and Multi-motor Drives - In Wheel Drives.
Braking system in electric and Hybrid vehicles, Regenerative braking, Hybrid braking.

CO3	**Outline the basic design considerations of EV**		
M3.01	List the design requirements for electric vehicles	4	Remembering
M3.02	Infer the dynamics and performance of electric vehicles during acceleration, rolling resistance ,aerodynamic drag ,vehicle moving up and down a hill	4	Remembering
M3.03	Illustrate the classification of motors used in EV	4	Understanding
M3.04	Summarize the comparison of motors used in EV	3	Understanding

Contents:

Design requirement for electric vehicles- Range, maximum velocity, acceleration, power

Requirement, mass of the vehicle. Various Resistances- rolling resistance , aerodynamic drag- Performance of electric vehicles acceleration, moving up and down a hill. Tractive Effort and Transmission. Requirement, Transmission efficiency, energy consumption.

Classification & Requirements of EV motors - Brushed & Brushless DC motors , Electric motor comparison

CO4	**Outline the braking system for EV's**		
M4.01	Outline the braking systems of EV	3	Understanding
M4.02	Illustrate regenerative braking and hybrid braking	4	Understanding

M4.03	Illustrate various sensors used in EV	4	Understanding
M4.04	Outline the service and maintenance of EV components	4	,Understanding
	Series Test II	1	

Contents:

Brake system for EV's - Air Conditioning Systems, Adaptive cruise control, ABS braking system

Cooling Systems - EV batteries and motors

Sensors used in EV's- Temperature sensor , MEMS, current sensors , coolant level sensor

Maintenance and safety of EV: Electric vehicle maintenance as compared to combustion engine , maintenance of electric motor, braking system ,battery ,sensors, electronic stability system.

Text/Reference:

T/R	Book Title/Author
R1	Jack Erjavec, Jeff Arias, "Hybrid, Electric and Fuel-cell Vehicles", Cengage Learning India Pvt Ltd. New Delhi.
R2	S. Onori, L. Serrao and G. Rizzoni, "Hybrid Electric Vehicles: Energy Management Strategies", Springer, 2015
R3	T. Denton, "Electric and Hybrid Vehicles", Routledge
R4	A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi (Ed.
R5	C. Mi, M. A. Masrur and D. W. Gao, "Hybrid Electric Vehicles: Principles and

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/108/103/108103009/
2	https://nptel.ac.in/courses/108/106/108106170
3	https://www.sciencedirect.com/science/article/pii/S2095756420300647
4	https://www.metisengineering.com
6	https://onlinecourses.nptel.ac.in/noc20_ee99/preview